MGM University

Chhatrapati Sambhajinagar

Name of Faculty – Engineering & Technology

Name of College – School of Engineering and Technology

Name of Department – Common for All Programs

Name of Programme – B.Tech (Integrated) First Year

MGMUNIVERSITY

CURRICULUM BOOKLET

(According to NEP 2020 with effect from Academic Year 2025-26)

MGM University

Chhatrapati Sambhajinagar

MGMUNIVERSITY

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MGMUNIVERSITY

MGM University

Chhatrapati Sambhajinagar

Vision

- To ensure stainable human development which encourages self-reliant and self-content society.
- To promote activities related to community services, social welfare and also Indian heritage and culture.
- To inculcate the culture of non-violence and truthfulness through vipassanna meditation and GandhianPhilosophy.
- To develop the culture of simpl eliving and high thinking.

Mission

- To impart state of art education and technical expertise to students and give necessary trainingto teachers to create self-reliant society for future.
- To encourage students to participate in Indian and International activities in sports, literature, etc. so that future generation becomes base for free and liberal society
- To educate students in areas like Management, Finance, Human relationstoin culcate philosophy of simple living and high thinking value of simple economic society.
- ToinculcatecultureofnonviolenceandtruthfulnessthroughVipassana.
- To sustain activities of Indian culture (viz.classical dance, music and fine arts) through establishing institutes like Mahagami, Naturopathy, etc.



विद्यापीठ गीत

अत्तदिपभवभवप्रदिपभव,
स्वरूपरूपभवहो
ज्ञानसब्बववज्ञानसब्बभव ,
सब्बदिपभवहो
अत्तादहअत्तनोनाथो ,
अत्तादहअत्तनोगतत
अत्तमागगपरअप्रमाि सेहैतुझेचलना
सब्बकाकल्याणहो ,
वोकायगकुशलकरना
सब्बकाउत्तममंगल ,पथप्रिशगकहो
अत्तदिपभवभवप्रदिपभव ,
स्वरूपरूपभवहो
ज्ञानसब्बववज्ञानसब्बभव ,
सब्बदिपभवहो
बुद्धमंशरनंगच्छामम :
धम्मंशरनंगच्छामम :
संघंशरनंगच्छामम :

School of Engineering & Technology (SOET) at a Glance

The School of Engineering and Technology (SoET), a self-aided, non-minority, constituent institution of MGM University is the first institution to provide integrated B.Tech (Integrated) and B.TechPrograms in the Marathwada region of Maharashtra. A sister institution of the illustrious Jawaharlal Nehru Engineering College (JNEC), Aurangabad, it was established as MGM Polytechnic College in 2010.

However, recognizing the rising need of quality engineering education after schooling in the region and beyond, it was transformed into the School of Engineering and Technology (SoET) in 2022. Inheriting a 42 years rich legacy of imparting world-class professional education from MGM Trust, SoET provides ultramodern research and innovation facilities to its students. Its integrated programs eliminate the need to appear for 12" standard and various engineering entrance tests, thus saving valuable time and energy of the students. It allows the students to utilize the same time to gain more practical knowledge of the field and a strong edge over the competition.

The institute offers a wide variety of Undergraduate (Six Years Integrated Programs after SSC), Undergraduate (Four Years Programs after HSC) programs to fulfill the dreams of the students. All these are advanced engineering programs founded on industry-oriented curriculum

The Institution employs many modern teaching-learning methods like Project-based Learning, Research-based Learning and Collaborative Learning to develop comprehensive understanding and real-life skills in the students. The School has also signed several memoranda of understanding with renowned institutions, associations and industries to provide internship and project work opportunities to the students. It also joined hands with NPTEL, SWAYAM, ATAL and other MOOC platforms to facilitate blended learning facilities for the students. Institute is unique in its structure, methods and goals, the institution is strongly rooted in the philosophy of training and research that enhances the relationship between knowledge and its application and seeks to promote the creation ofan ideal society through holistic education

Vision

Enrich the students in technical education to work globally with human values.

Mission

- To inculcate technical skills by continuous assessment practices.
- To strengthen interaction between industry and institute.
- To involve the students in social activities with environmental awareness.

Programs offered at SOET

B.Tech (Integrated) UG Program 6 Years (after 10th)	**B.Tech UG Program 4 Years (after 12th)**
Civil Engineering	Computer Science and Engineering (IoT, Cyber Security Including Block Chain Technology)
Computer Science and Engineering (Data Science)	Computer Science and Engineering (Cyber Security)
Computer Science and Engineering (IoT, Cyber Security Including Block Chain Technology)	Electrical Engineering
Robotics and AI	Electronics Engineering (VLSI Design and Technology)
--	Robotics and Automation

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech (Integrated) All Programs

Programme Type (UG/PG): UG

Duration: 06 Years

Semester - I

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs / Week)		Evaluation Scheme (Marks)			Minimum Passing(Marks)		
					L	P	Internal	External	Total	Internal	External	Total
BSC	APS24BSL101	Basic Mathematics	Lecture	4	4	-	60	40	100	-	16	40
BSC	APS24BSL102	Physics - I	Lecture	2	2	-	60	40	100	-	16	40
BSC	APS24BSL103	Chemistry - I	Lecture	2	2	-	60	40	100	-	16	40
AEC	APS24AEL104	Basic Communication Skills	Lecture	2	2	-	60	40	100	-	16	40
ESC	APS24ESL105	Introduction to Engineering	Lecture	3	3	-	60	40	100	-	16	40
BSC	APS24BSP101	Basic Mathematics Lab	Practical	1	-	2	30	20	50	-	8	20
BSC	APS24BSP102	Physics - I Lab	Practical	1	-	2	30	20	50	-	8	20
BSC	APS24BSP103	Chemistry - I Lab	Practical	1	-	2	30	20	50	-	8	20
AEC	APS24AEP104	Basic Communication Skill Lab	Practical	2	-	4	60	40	100	-	16	40
VSEC	APS24VSP106	Information and Communications Technology Lab	Practical	2	-	4	60	40	100	-	16	40
TOTAL				**20**	**13**	**14**	**510**	**340**	**850**	-	-	-

Nature of Course: L-Lecture, P-Practical

Course Category: BSC – BasicScienceCourse, ESC – EngineeringScienceCourse, AEC – AbilityEnhancement Course, PCC – Program Core Course, VSEC - Vocational Skill and Skill EnhancementCourse, CCA-Co-CurricularActivity

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech (Integrated) All Programs

Programme Type (UG/PG): UG

Duration: 06 Years

Semester - II

Course Category	Course Code	Course Title	Nature of Course	No. of Credits	Teaching (Contact Hrs / Week)		Evaluation Scheme (Marks)			Minimum Passing(Marks)		
					L	P	Internal	External	Total	Internal	External	Total
BSC	APS24BSL107	Engineering Mathematics	Lecture	3	3	-	60	40	100	-	16	40
BSC	APS24BSL108	Physics - II	Lecture	3	3	-	60	40	100	-	16	40
BSC	APS24BSL109	Chemistry - II	Lecture	2	2	-	60	40	100	-	16	40
VEC	APS24VEL110	Universal Human Values	Lecture	2	2	-	30	20	50	-	8	20
ESC	APS24ESL111	Elements of Mechanical Engineering	Lecture	2	2	-	60	40	100	-	16	40
BSC	APS24BSP107	Engineering Mathematics Lab	Practical	1	-	2	30	20	50	-	8	20
BSC	APS24BSP108	Physics - II Lab	Practical	1	-	2	30	20	50	-	8	20
BSC	APS24BSP109	Chemistry - II Lab	Practical	1	-	2	30	20	50	-	8	20
ESC	APS24ESP111	Elements of Mechanical Engineering Lab	Practical	1	-	2	30	20	50	-	8	20
VSEC	APS24VSP112	Web Development Lab	Practical	2	-	4	60	40	100	-	16	40
VSEC	APS24VSP113	Engineering Workshop - I	Practical	2	-	4	60	40	100	-	16	40
TOTAL				**20**	**12**	**16**	**510**	**340**	**850**	-	-	-

Nature of Course: L-Lecture, P-Practical

Course Category: BSC – Basic Science Course, ESC – Engineering Science Course, AEC – Ability Enhancement Course, PCC – Program Core Course, VSEC - Vocational Skill and Skill Enhancement Course, CCA-Co-Curricular Activity

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech (Integrated) All Programs

Programme Type (UG/PG): UG

Duration: 06 Years

Semester - III												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact Hrs / Week)**		**Evaluation Scheme (Marks)**			**Minimum Passing(Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
BSC	APS24BSL201	Mathematics - III	Lecture	4	4	-	60	40	100	-	16	40
BSC	APS24BSL202	Physics – III	Lecture	3	3	-	60	40	100	-	16	40
PCC	APS24PCL203	Introduction to AI & Data Science	Lecture	2	2	-	60	40	100	-	16	40
ESC	APS24ESL204	Basics of Electrical & Electronics Engineering	Lecture	4	4	-	60	40	100	-	16	40
PCC	APS24PCL205	Programming in C Language	Lecture	2	2	-	60	40	100	-	16	40
BSC	APS24BSP202	Physics - III Lab	Practical	1	-	2	30	20	50	-	8	20
ESC	APS24ESP204	Basics of Electrical & Electronics Engineering Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	APS24PCP205	Programming in C Language Lab	Practical	1	-	2	30	20	50	-	8	20
VEC	APS24VEP206	Social and Life Skill Lab	Practical	1	-	2	30	20	50	-	8	20
VSEC	APS24VSP207	Engineering Workshop - II	Practical	1	-	2	30	20	50	-	8	20
TOTAL				**20**	**15**	**10**	**450**	**300**	**750**	-	-	-

Nature of Course: L-Lecture, P-Practical

Course Category: BSC – Basic Science Course, ESC – Engineering Science Course, AEC – Ability Enhancement Course, PCC – Program Core Course, VSEC - Vocational Skill and Skill EnhancementCourse, CCA - Co-Curricular Activity

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B.Tech (Integrated) All Programs

Programme Type (UG/PG): UG

Duration: 06 Years

Semester - IV													
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact Hrs / Week)**		**Evaluation Scheme (Marks)**			**Minimum Passing(Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
BSC	APS24BSL208	Mathematics - IV	Lecture	4	4	-	60	40	100	-	16	40
ESC	APS24ESL209	Basics of Engineering Drawing	Lecture	3	3	-	60	40	100	-	16	40
ESC	APS24ESL210	Introduction to Robotics	Lecture	2	2	-	60	40	100	-	16	40
ESC	APS24ESL211	Basics of Civil Engineering	Lecture	2	2	-	60	40	100	-	16	40
PCC	APS24PCL212	Introduction to Cyber Security & Blockchain	Lecture	2	2	-	60	40	100	-	16	40
ESC	APS24ESP209	Basics of Engineering Drawing Lab	Practical	2	-	4	60	40	100	-	16	40
ESC	APS24ESP210	Introduction to Robotics Lab	Practical	1	-	2	30	20	50	-	8	20
ESC	APS24ESP211	Basics of Civil Engineering Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	APS24PCP213	Fundamentals of IOT Lab	Practical	1	-	2	30	20	50	-	8	20
VSEC	APS24VSP214	Computer Aided Drafting Lab	Practical	2	-	4	60	40	100	-	16	40
TOTAL				**20**	**13**	**14**	**510**	**340**	**850**	-	-	-

Nature of Course: L-Lecture, P-Practical

Course Category: BSC – Basic Science Course, ESC – Engineering Science Course, AEC – Ability Enhancement Course, PCC – Program Core Course, VSEC - Vocational Skill and Skill Enhancement Course, CCA-Co-Curricular Activity

Syllabus

Semester–I

Course Code: APS24BSL101
Course Name: Basic Mathematics
Course Category: BSC
Credits: 4
Teaching Scheme: L-4
Evaluation Scheme: CA:60, ESE:40
Duration of ESE: 3 Hours
Pre-requisites: Basic Concept of Algebra, Elementary Mathematics
Course Objectives: 1. To Understand Basic of algebra , Determinant , Matrices and Partial fractions 2. To Understand the different trigonometric ratios and formulae and applications. 3. To Understand Straight line and their properties 4. To Understand Line and Plane 5. To Understand Scalar and Vectors
Course Outcomes: At the end of the course, the students will be able to – **CO1:** Apply the concepts of algebra to solve engineering (discipline) related problems **CO2:** Utilize trigonometry to solve branch specific engineering problems **CO3:** Solve area specific engineering problems under given conditions of straight lines **CO4:**Understand Plane and its properties **CO5**: Apply vector operation to solve engineering applications.

Contents:

Unit No.	Content	Teaching Hours
1.	**Algebra** 1.1 Determinant : concept and examples of second and third order 1.2 Matrices: Matrices, algebra of matrices, transpose, value of determinant of matrix of order 3x3, adjoint and inverse of matrices. Matrices: Solution of simultaneous equations by matrix inversion method. 1.3 Partial Fractions: Types of partial fraction based on nature of factors and related Problems.	12
2.	**Trigonometry** 2.1 Trigonometric ratios of allied angles, compound angles, multiple angles, submultiples angles. 2.2 Factorization and Defactorization formulae. 2.3 Inverse Trigonometric Ratios and related problems. 2.4 Principle values and relation between trigonometric and inverse trigonometric ratios.	14
3.	**Straight Line** 3.1 Slope of straight line, Angle between two lines, Condition of parallel and perpendicular lines. 3.2 Various forms of lines 3.3 Perpendicular distance from a point on the line. 3.4 Perpendicular distance between two parallel lines.	12
4.	**Line and Plane** 4.1 Concept of Plane, Angle between two intersecting planes, Angle between	12

	line and plane, condition for perpendicularity and parallelness of planes 4.2 Equation of planes in different forms.	
5.	**Vectors** 5.1 Scalars and Vectors, Algebra of vectors 5.2 Vectors of 2 & 3 dimensions, scalar and vector product 5.3 Applications of vectors to area of triangle, work done by a force 5.4 Scalar and vector triple product and applications	10

Text Books:

1.Hall and Knight, Higher Algebra ArihantPublicationsIndialimited,
2.Loney,SL(1912),Plane Trigonometry,Cambridge:UniversityPress,JFM25.0892.12
3.Loney,SL(1895),The elements of coordinate geometry, London:MacmillanandCo

Reference Books:

1 Grewal B. S.; Higher Engineering Mathematics; Khanna publication NewDelhi,2013 ISBN: 8174091955
2. Das H.K.; Advance Engineering Mathematics : S Chand publication New Delhi 2008 ISBN: 9788121903455
3. Kreysizg, Ervin ; Advance Engineering Mathematics : Wiley publication New Delhi 2016 ISBN: 978-81-265-5423-2

ICT / MOOCS: NPTEL / SWAYAM Lectures

Course Code: APS24BSL101
Course Name: Basic Mathematics Lab
Course Category: BSC
Credits: 1
Teaching Scheme: P-2
Evaluation Scheme: TW:30, PR:20
Pre-requisites: Basic of algebra, Trigonometry, Straight line, Plane, Vectors
Course Objectives: 1. To Understand Basic of algebra , Determinant , Matrices and Partial fractions 2. To Understand the different trigonometric ratios and formulae and applications. 3. To Understand Straight line and their properties 4. To Understand Line and Plane 5. To Understand Scalar and Vectors
Course Outcomes: At the end of the course, the students will be able to – **CO1:** Apply the concepts of algebra to solve engineering (discipline) related problems **CO2:** Utilize trigonometry to solve branch specific engineering problems **CO3:** Solve area specific engineering problems under given conditions of straight lines **CO4:** Understand Plane and its properties **CO5**: Apply vector operation to solve engineering applications.

Contents –

Sr. No.	List of Experiments	Lab Hours
1.	Solve simple problems Determinant of order 2 and 3.	2
2.	Solve elementary problems on Algebra of matrices for branch specific engineering related applications.	2
3.	Resolve into partial fraction using linear non- repeated, repeated, and irreducible quadratic factors.	2
4.	Solve problems on Compound, Allied, multiple and submultiple angles for related shapes.	2
5.	Practice problem son factorization and de factorization.	2
6.	Solve problems on inverse trigonometric ratios based on applications.	2
7.	Practice problems on equation of straight lines using different forms.	2
8.	Solve problems on perpendicular distance, distance between two parallel lines and angle between two lines.	2
9.	Solve problems on angle between Plane and Angle between line and Plane.	2
10.	Practice Perpendicularity and Parallelness of plane.	2

11.	Solve problems on Algebra of Vectors.	2
12.	Application of vectors to find area and work done.	2

LABORATORY EQUIPMENT/INSTRUMENTS/TOOLS/SOFTWAREREQUIRED

Sr.No	Equipment Name with Broad Specifications
1	Open-source software like SageMaths, MATHS3D, GeoGebra, Graph, DPLOT, for straight line, Plane, Vectors and Graphing Calculator can be used for Algebra, Trigonometry, respectively.

Course Code: APS24BSL102
Course Name: Physics - I
Course Category: BSC
Credits: 2
Teaching Scheme: L-2
Evaluation Scheme: CA:60, ESE:40
Duration of ESE: 2 Hours
Pre-requisites:
Course Objectives: 1. To understand the units of various physical quantities. 2. To understand the concept of magnetic materials and semiconductor devices. 3. To understand the concept of thermodynamics with engineering applications. 4. To understand the concept of light phenomena and applications of optical fibers.
Course Outcomes: At the end of the course, the students will be able to – **CO1:** Use basic devices to measure the physical quantities in various engineering situations. **CO2:** Apply the basic principles of electromagnetism to solve given engineering problems. **CO3:** Apply basic principles of thermodynamics to solve engineering problems. **CO4:** Apply basic principles of fiber optics to solve engineering problems.

Contents:

Unit No.	Content	Teaching Hours
1.	**Unit and measurements** 1.1 Unit, physical quantities: fundamental and derived quantities and their units Systems of units: CGS, MKS and SI. 1.2 Scalar and Vector Physical Quantities. 1.3 Dimensions, dimensional formula, Applications of dimensional analysis; correctness of physical equations, conversion factor for inter conversion of units in different systems of units. 1.4 Errors, types of errors: instrumental, systematic and random error, estimation of errors: absolute, relative and percentage error, significant figures.	8
2.	**Electromagnetism and Semiconductor** 2.1 Concept of charge, Coulomb's inverse square law, Electric field, Electric field intensity, potential and potential difference. 2.2 Magnetic field and magnetic field intensity and its units, magnetic lines of force, magnetic flux. 2.3 Electric current, Ohm's law, specific resistance, laws of series and parallel combination of resistance, Heating effect of electric current. 2.4 Conductors, Insulators and Semiconductors, Energy bands, intrinsic and extrinsic semiconductors, p-n junction diode, I-V characteristics of p-n junction, applications of p-n junction diode	8
3.	**Thermodynamics** 3.1 Heat, temperature, temperature scale: Celsius, Kelvin, Fahrenheit. 3.2 Modes of heat transfer: Conduction, Convection and Radiation, Applications in daily life. 3.3 Boyle's law, Charle's law, Gay Lussac's law, perfect gas statements equations and simple numerical. 3.4 Law of thermal conductivity, Newton's law of cooling.	7
4.	**Optics**	7

	4.1 Reflection, Refraction , Laws of reflection, Law of refraction, Snell's law. 4.2 Refractive index of prism, Total internal reflection, Optical fiber: Principle, construction and working 4.3 Types of Optical fibers, Single mode & Multimode step index, Applications of optical fibers 4.4 Production of X-rays by modern Coolidge tube, properties and engineering applications.	

Text Books:

1.Physics Textbook Part I - Class XI Narlikar J. V. Joshi, A. W. Mathur, Anuradha
2.Physics Textbook Part II - Class XI Narlikar, J.V. Joshi, A. W. Mathur, Anuradha National Council of Education Research and Training, New Delhi, 2015,ISBN:8174505660
3.Physics Textbook Part I - Class XII Narlikar J.V. Joshi , A. W. Ghatak A.K. National Council of Education Research and Training, New Delhi,2013, ISBN : 8174506314
4.Physics Textbook Part II - Class XII Narlikar J.V. Joshi , A. W. Ghatak A.K.

Reference Books:

1.Engineering Physics R.K.Gaur& S.K. Gupta DhanpatRai Publication- New Delhi
2. Applied physics –II Dr. HussainJeeva Khan
3.Engineering Physics M.N. Avadhanula P.G. Kshirsagar , S. Chand Company Ltd- New Delhi
4. Fundamental of Physics Jearl Walker Willey India Edition

ICT / MOOCS: NPTEL / SWAYAM Lectures

1. https://nptel.ac.in/courses/115106090
2. https://onlinecourses.swayam2.ac.in/nce19_sc05/preview
3. https://iitpal.iitd.ac.in/topic_list/2

Course Code: APS24BSP102
Course Name: Physics - I Lab
Course Category: BSC
Credits: 1
Teaching Scheme: P-2
Evaluation Scheme: TW:30, PR:20
Pre-requisites:
Course Objectives:
1. To understand the units of various physical quantities.
2. To understand the concept of magnetic materials and semiconductor devices.
3. To understand the concept of thermodynamics with engineering applications.
4. To understand the concept of light phenomena and application of optical fibres.
Course Outcomes: At the end of the course, the students will be able to –
CO1:.Use basic devices to measure the physical quantities in various engineering situations.
CO2: Apply the basic principles of electromagnetism to solve given engineering problems.
CO3: Apply the basic principles of thermodynamics to solve engineering problems.
CO4: Apply the basic principles of fibre optics to solve engineering problems.

Contents –

Sr. No.	List of Experiments	Lab Hours
1.	Measurements of dimensions of given object by Vernier calliper.	2
2.	Measurements of dimensions of given objects by micrometer screw gauge.	2
3.	Determination of resistance by Ohm's law.	2
4.	Determination of specific resistance of given wire.	2
5.	Determination of static and dynamic resistance of given P N junction diode.	2
6.	Determination of neutral points by magnetic compass.	2
7.	Determination of forbidden energy band gap in semiconductors.	2
8.	Determination of Joule's mechanical equivalent of heat by Joule's law	2
9.	Determination of Coefficient of thermal conductivity.	2
10.	Determination of the refractive index of glass slab	2

Text Books: 1.Physics Textbook Part I - Class XI Narlikar J. V. Joshi, A. W. Mathur, Anuradha 2.Physics Textbook Part II - Class XI Narlikar, J.V. Joshi, A. W. Mathur, Anuradha National Council of Education Research and Training, New Delhi, 2015,ISBN:8174505660 3.Physics Textbook Part I - Class XII Narlikar J.V. Joshi , A. W. Ghatak A.K. National Council of Education Research and Training, New Delhi,2013, ISBN : 8174506314 4.Physics Textbook Part II - Class XII Narlikar J.V. Joshi , A. W. Ghatak A.K.
Reference Books: 1.Engineering Physics R.K.Gaur& S.K. Gupta DhanpatRai Publication- New Delhi 2. Applied physics –II Dr. HussainJeeva Khan 3.Engineering Physics M.N. Avadhanula P.G. Kshirsagar , S. Chand Company Ltd- New Delhi 4. Fundamental of Physics Jearl Walker Willey India Edition
ICT / MOOCS: NPTEL / SWAYAM Lectures 1.https://nptel.ac.in/courses/115106090 2.https://onlinecourses.swayam2.ac.in/nce19_sc05/preview 3.https://iitpal.iitd.ac.in/topic_list/2

Course Code:APS24BSL103
Course Name: Chemistry - I
Course Category: BSC
Credits: 2
Teaching Scheme: L-2
EvaluationScheme: CA:60, ESE:40
Duration of ESE: 2 Hours
Pre-requisites: Basic Engineering Chemistry.
Course Objectives: 1. To provide students with a functional understanding of structure, properties and reaction of organic compounds. 2. To provide students with functional understanding of structure of atoms and how they bond together to fomr molecules and compounds. 3. Equip students with function understanding of this materials and there properties and applications. 4. Equip studnest with fundamental understanding of principle govering the interaction between chemiscal and electrical energy.
Course Outcomes: At the end of the course, the students will be able to – CO1**:** Demonstrate an understanding of basic principles of organic chemistry and how they relate to everyday experiences. CO2: Illustrate the Basic Concept of Structure of Atom CO3: Select the Relevant metallurgic Process related to Industrial Applications CO4: Use Relevant Electrolyte in Electrolysis for different Applications

Contents:

Unit No.	Content	Teaching Hours
1.	**Basic Principles of Organic Chemistry :** 1.1. Introduction, 1.2 Structural representation of organic molecules-Condensed formula, structural formula, Bond line or Zig-zag formula with examples. 1.3. Classification of organic compound. Acyclic/Open chain compound. Cyclic compound, Homocyclic,Heterocycliccompound,Alicyclic,Aromatic (Benzenoid, Nonbenzenoid),Heterocyclic (Aromatic . Nonaromatic),(Definition with example.) 1.4. Functional group in organic compounds. Halides,Cyanides,Isocyanides, Nitro compound ,Alcohol,Phenol,Amines.Ether,Aldehydes,Ketones,Carboxylic acid, Ester, Amide, Sulphonic acid, Acyclic chloride,Acid unhydride,Alkyne,Alkene,Alkane.(Structure and Functional group with examples).	08
2.	**Atomic Structure and Chemical bonding** 2.1 Introduction 2.2• Definition of atom, Bohr's atomic model, definition of atomic number, atomic mass number and their differences, 2.3 Isotopes and Isobars: Definitions, examples and distinction 2.4 Orbits and orbitals: s, p, d, f orbitals, shapes and description of s-orbital	07

	and p- orbital. Distribution of electrons in orbitals: 2.5 Definition of electronic configuration, Aufbau's principle, Hund's rule, orbital Electronic configurations (s, p, d, f) of elements having atomic number 1 to 30,10 2.6 Valance Definitions of valence. Definition of electrovalence, positive and negative electrovalence, formation of Electrovalent compounds-MgO,CaCl2 Definition of covalence, single, double and triple covalent bonds, formation of Covalent compounds H2O,CO2,N2.	
3	**Metals and Alloys** 3.1 Introduction 3.2 Occurrence of metals in free and combined state, definitions-mineral, ore, Gangue, flux and slag, metallurgy. 3.3 Mechanical properties ofmetals- Hardness, ductility, malleability, tensile strength, toughness, machinability, weld- ability, forging, soldering, brazing, cast ability. 3.4.Alloys Definition, purposes of making alloys with examples. Preparation methods- Fusion, Compression Classification of Alloys- Ferrous and nonferrous alloys with examples. Examples of alloys-Composition, properties and application of duralumin, woods metal.	08
4	**Electrochemistry :** 4.1 Introduction 4.2 Basic concepts of electrolysis: Electrolyte, types of electrolyte- strong and weak electrolyte, their Difference. 4.3 Ionization and electrolytic dissociation, Arrhenius theory of electrolytic Dissociation, degree of ionization, factors affecting degree of ionization. 4.4Definitions of electrolytic cell, electrodes-cathode, anode, electrode Potential-oxidation potential and reduction potential. 4.5Electrolysis: Mechanism of electrolysis- Electrolysis, electrochemical series for cations and anions, Mechanism of electrolysis of CuSO4 solution by using platinum electrodes and copper electrodes 4.6 Applications of electrolysis- Electroplating of silver, electro refining of blister copper, 4.7 Faraday's laws of electrolysis: Faraday's first and second law, relation between electrochemical equivalent and chemical equivalent. 4.8. pH and pOH: Definition of pH, pOH, pH Scale	07

Text Books:

1. Jain P.C & Jain Monica, Engineering Chemistry, DhanpatRai& Sons, Delhi
2. Bhal&Tuli, Text book of Physical Chemistry, S. Chand & Company, New Delhi.
3. O. G. Palanna, Engineering Chemistry, Tata McGraw-Hill Publication, New Delhi.
4. S. S. Dara, A textbook of Engineering Chemistry, McGraw-Hill Publication, New Delhi.
5. ShashiChawla, Engineering Chemistry, DhanpatRai and Co Education and Technical Publishers
6. V. R Gowarikar, Polymer Science, New Age International Ltd.

7. B. K. Sharma, Instrumental Methods of Chemical Analysis, Krishna Prakashan Media

Reference Books:

1. Barrow G.M., Physical Chemistry, McGraw-Hill Publication, New Delhi.
2. ShikhaAgrawal, Engineering Chemistry- Fundamentals and Applications, Cambridge Publishers.
3. Atkins, Physical Chemistry, OUP Oxford.
4. Engineering Chemistry, Wiley India Pvt, Ltd, First edition

E-Resources: Name of the website/ E-Journals/ Online Videos

1. NPTEL Basic Courses Engineering Chemistry (https://nptel.ac.in/courses/122/101/122101001/)
2. https://www.ncertbooks.guru/engineering-chemistry/
3. Coursera Chemistry Courses (https://www.coursera.org/browse/physical-science-and-engineering/chemistry?languages=en&page=2)
4. "Introduction to Corrosion" NPTEL Course (https://www.youtube.com/playlist?list=PL8lylDWRkaW8BXestE4baRnN0699S11Lq)
5. "Tribology & Lubrication" NPTEL Course (https://www.youtube.com/playlist?list=PLLwnvFq-JAltJvWAfEVU4gcUK27hEA7FD)
6. "Electrochemistry" NPTEL Course (https://www.youtube.com/playlist?list=PLVFqK_9GOGXnnriQpsn0z1Rss96Rh0vsm)
7. "Modern Instrumental Methods of Analysis" NPTEL Course (https://www.youtube.com/playlist?list=PL400CAFBA72E94CF8)

Course Code: APS24AEP103
Course Name: Chemistry - I Lab
Course Category: BSC
Credits: 1
Teaching Scheme: P-2
Evaluation Scheme: TW:30, PR:20
Pre-requisites: Basic Engineering Chemistry.
Course Objectives:
1. Demonstrate an understanding of basic principles of organic chemistry and how they relate to everyday experiences.
2. Illustrate the Basic Concept of Structure of Atom.
3. Select the Relevant metallurgic Process related to Industrial Applications.
4. Use Relevant Electrolyte in Electrolysis for different Applications.
Course Outcomes: At the end of the course, the students will be able to –
LO1:.Design and carry out experiments as well as accurately record and analyse the result of experiments.
LO2: Illustrate the Basic Concept of Structure of Atom.
LO3: The Relevant metallurgic Process related to Industrial Applications.
LO4: Relevant Electrolyte in Electrolysis for different Applications.

Contents –

Sr. No.	List of Experiments	Lab Hours
1.	Do you know the Chemistry Laborotary.	2
2.	Determine preliminary test and functional group of organic comp.1	2
3.	Determine preliminary test and functional group of organic comp.2	2
4.	Determine preliminary test and functional group of organic comp.3	2
5.	Determine preliminary test and functional group of organic comp.4	2
6.	Identify an ion in the given ionic solution (non metallic ion).	2
7.	Identify an ion in the given ionic solution (metallic ion).	2
8.	Determination of acid value of an oil sample.	2
9.	Determine the percentage of Iron in the given sample using redox titration.	2
10.	Determine the pH value of given solution using universal indicator.	2
11	Determine the voltage generated from chemical raction using Danial cell.	2
12	Determine the electrode potential of iron metal.	2

Reference Books:

1. A Text book on Experiments and Calculations in Engineering Chemistry Author Dr. S. S. Dara, S Chand Publication.
2. Experiments in Engineering Chemistry by Payal B. Joshi, I K International Publishing House Ltd.
3. Systematic Experimental Physical Chemistry by S. W. Rajbhoj& T. K. Chondekar, Anjali Publication.
4. Practical Chemistry with Experiments by Newell, Lymam C.
5. Text Book of Macro and Semi micro inorganic analysis by Vogel. Longman group Ltd London.
6. Engineering Chemistry with Laboratory Experiments by M. S. Kaurav, Asia-Pacific the holdings Private Ltd.

Course Code:APS24AEL104
Course Name: Basic Communication Skills
Course Category: AEC
Credits: 2
Teaching Scheme: L-2
Evaluation Scheme: CA:60, ESE:40
Duration of ESE: 2 Hours
Pre-requisites: Basic knowledge of English language : Listening, speaking, reading & writing
Course Objectives: 1. To make students aware of basics of communication in day to day situations 2. To enable students know the importance of non verbal communication. 3. To make students understand basic sounds of English. 4. To enable students evaluate the different communicative style of individuals.
Course Outcomes: At the end of the course, the students will be able to – **CO1:** understand basic concepts and principles of communication. **CO2:** demonstrate effective use of body language in conversations. **CO3:** use appropriate words and their pronunciations as per the context. **CO4:** deliver prepared speeches to express ideas, emotions and thoughts.

Contents:

Unit No.	Content	Teaching Hours
1.	**Nature of Communication:** 1.1 Introduction to communication 1.2 Communication process and its elements 1.3 Types of communication 1.4 Basic principles of communication 1.5 Barriers to communication	8
2.	**Verbal and Non-verbal communication:** 2.1 Verbal communication: oral and written 2.2 Non verbal communication: Body language 2.3 Non-verbal codes 2.4 Grooming etiquettes for formal situations.	7
3.	**Introduction to the sounds of English and Speaking Skills:** 3.1 English Phonetics: Introduction, Organs of Speech 3.2 Consonant and vowel sounds 3.3 Accent, rhythm and intonation. 3.4 Speaking skills	8
4.	**Speech Writing:** 4.1 Self Introduction 4.2 Welcome Speech, 4.3 Farewell Speech, 4.4 Introducing a Guest and Vote of Thanks	7

Text Books: 1. Text book of Communication Skills. (Sanjay Kumar, PushpaLata) Oxford University Press. 2.Communication Skills (URMILA RAI) Himalaya Publications 3. English for success (E.Sureshkumar, P. Sreehari, and J.Savithri) Foundation Books. 4.Effective communication skills (M.Ashraf Rizvi) 5.Better English pronunciation
Reference Books: 1. English for success (E.Sureshkumar, P. Sreehari, and J.Savithri) Foundation Books**.** 2.Effective communication skills (M.AshrafRizvi)
ICT / MOOCS: NPTEL / SWAYAM Lectures

Course Code: APS24AEL104
Course Name: Basic Communication Skills Lab
Course Category: AEC
Credits: 2
Teaching Scheme: P-2
Evaluation Scheme: TW: 60, PR: 40
Pre-requisites: Basic knowledge of English language: Listening, speaking, reading & writing
Course Objectives:
1. Assimilate basic concepts of formal communication.
2. Enhance students' competency to effectively use body language in professional settings.
3. Introduce sounds of English.
4. Apply social skills in day to day life.
Course Outcomes: At the end of the course, the students will be able to –
CO1: understand basic concepts and principles of communication
CO2: demonstrate effective use of body language in conversations.
CO3: use appropriate words and their pronunciations as per the context
CO4: deliver prepared speeches to express ideas, emotions and thoughts

Contents –

Sr. No.	List of Experiments	Lab Hours
1.	Self-introduction	2
2.	Formal and Informal communication	2
3.	Non-verbal communication.	2
4.	Barriers to communication	2
5.	Interview techniques	2
6.	Introduction to phonetics	2
7.	Sound system in English language (vowels and consonants)	2
8.	Presentation Strategies	2
9.	Speeches on the given formal situations	2
10.	Interpersonal skills in public sphere	2

Text Books: 1. Text book of Communication Skills. (Sanjay Kumar, PushpaLata) Oxford University Press. 2.Communication Skills (URMILA RAI) Himalaya Publications 3. English for success (E.Sureshkumar, P. Sreehari, and J.Savithri) Foundation Books. 4.Effective communication skills (M.AshrafRizvi) 5.Better English pronunciation
Reference Books: 1. English for success (E.Sureshkumar, P. Sreehari, J.Savithri) Foundation Books**.** 2.Effective communication skills (M.Ashraf Rizvi)
ICT / MOOCS: NPTEL / SWAYAM Lectures

Course Code:APS24ESL105
Course Name: Introduction to Engineering
Course Category: ESC
Credits: 3
Teaching Scheme: L-3
Evaluation Scheme: CA:60, ESE:40
Duration of ESE: 2 Hours
Pre-requisites:
Course Objectives: 1. Introduce the scope, applications, and significance of each discipline in the context of national development and technological advancement. 2. Familiarize students with essential concepts, materials, components, and processes specific to each branch. 3. Develop a broad interdisciplinary perspective by highlighting modern trends such as smart infrastructure, embedded systems, cyber security, renewable energy, and robotics. 4. Build foundational technical vocabulary and conceptual knowledge to support further specialized studies in their respective engineering fields. 5. Encourage awareness of the roles and responsibilities of engineers in shaping infrastructure, industries, and digital ecosystems.
Course Outcomes: At the end of the course, the students will be able to – **CO1:** Understand the Scope and Role of Civil Engineering **CO2:** Understand the Scope and Role of Computer Science Engineering **CO3:** Understand the Scope and Role of Electrical and Electronics Engineering **CO4:** Understand the Scope and Role of Mechanical Engineering

Contents:

Unit No.	Content	Teaching Hours
1.	**Introduction to Civil Engineering** Various Disciplines of Civil engineering, Importance of Civil Engineering in Infrastructure Development of the country, Introduction to Types of Buildings as per NBC, Fundamental Civil Engineering Materials & Techniques, Modern Trends in Civil Engineering.	12
2.	**Introduction to Computer Science Engineering** Definition, History, Components: CPU, RAM, Storage, Input & Output Devices, Motherboard, System Software Vs. Application Software, Introduction to Number System, Algorithm and Program, Roles in Computer Engineering, Software Development, Embedded Systems, Cyber Security.	11
3.	**Introduction to Electrical and Electronics Engineering** History and Scope of Electrical Engineering, Applications in Modern Technology , Basic Electrical Quantities: Voltage, Current, Power, and Energy, A brief history of Electronics, Definition of Semiconductor, Types of Semiconductor, Active and Passive Components Resistor, Inductor, Capacitor Symbol.	12
4.	**Introduction to Mechanical Engineering** Introduction to Thermal Engineering: Thermodynamics basic definition, System and Process, Modes of Heat Transfer, Heat Engines and Power plants. Introduction to Machines and Materials: Basics of Theory of Machines, Power Transmissions and Material Science, Heat treatment, Types of Manufacturing process and Introduction to Welding Process. Introduction to Robotics.	12

Text Books:

1. S.S. Bhavikatti, *Basic Civil Engineering*, New Age International Publishers.
2. B. Ram, Fundamentals of Computers.
3. S. K. Bhattacharya, Basic Electrical and Electronics Engineering, Pearson Education India.
4. P.K.Nag, Power Plant Engineering, McGraw Hill Education.

Reference Books:

1. Rangwala, *Engineering Materials*, Charotar Publishing House.
2. Basic Civil and Mechanical Engineering" by G. Shanmugam, M. S. Palanichamy, and S. Ravindran.
3. **National Building Code of India (NBC 2016)**, Bureau of Indian Standards.
4. B. Ram, *Fundamentals of Computers*.
5. S. Salivahan, Basic Electrical and Electronics Engineering, McGraw Hill Publicaiton.
6. P.K.Nag, *Power Plant Engineering*, McGraw Hill Education.

ICT / MOOCS: NPTEL / SWAYAM Lectures

Course Code:APS24VSP106
Course Name: Information and Communication Technology Lab
Course Category: VSEC
Credits: 2
Teaching scheme: P- 4
Evaluation Scheme: TW:60, PR: 40
Pre-requisites: Basic Computer Skills, Understanding of Digital Technologies
Course Objectives: The main aim of ICT is to help students to become competent and confident users who can use the basic knowledge and skills acquired to assist them in their daily lives. It is also supposed to prepare students for the world of tomorrow. It aims to help learners to have an open and flexible mind. This will help them to adjust to the inevitable future changes.
Course Outcomes: CO1: Explain the concept of ICT and its related terminologies. CO2: Describe the types of information systems. CO3: Demonstrate an understanding of the basic concepts and requirements for data communication and networking. CO4: Utilize the basic concepts and requirements learnt for internet connectivity. CO5:Demonstrate responsible use of ICTs

Contents

Sr. No.	List of Experiments	Practical Hours
1.	Introduction to Information and Communication Technology.	2
2.	Study of Computer and its Peripherals Devices.	2
3.	Study of Motherboard ,Processor and BIOS.	2
4.	Demonstrate the use of Input and Output Devices.	2
5.	Communication and Networking-Introduction to E-Mail , Creating an E-Mail Account, Receiving and Replying to E-Mail.	2
6.	Study of Hard Disk and MODEM.	2
7.	Create Edit and Save Document AND Use of Various File Formatting Options on Word Document.	2
8.	Use of Images and Shapes in Word Document.	2
9.	Use of Tables in Document.	2
10.	Create Open and Edit Worksheets.	2
11.	Working with Formulas and Function in Worksheet.	2
12.	Sort, Filter and Validation of Data.	2
13.	Use of Charts for Visual Representation.	2
14.	Study of Power Supply.	2

15.	Slide presentation using Table and Charts.	2
16.	Animation effects to Text and slides.	2
17.	Internet for different web services.	2
18.	Networking and Maintenance	2

Text Books:

1.**Introduction to Information and Communication Technology in Education"**

- **Author**: M. D. Roblyer, Aaron H. Doering
- **Overview**: This textbook provides a solid foundation in ICT, with a particular focus on education. It covers the basic concepts of ICT and their application in teaching and learning.

2.**"Computer Networks"**

- **Author**: Andrew S. Tanenbaum, David J. Wetherall
- **Overview**: This book is a classic in the field of networking and covers topics such as protocols, network architecture, TCP/IP, and internet working in great detail.

3."Fundamentals of Computers"

- **Author**: V. Rajaraman
- **Overview**: A comprehensive textbook for introductory courses in computer science, covering hardware, software, and basics of programming.

Reference Books:

1.**"The Art of Computer Programming"**

- **Author**: Donald E. Knuth
- **Overview**: This is a series of books that dives deep into algorithms and the theory behind computer programming. It's a great reference for understanding the complexity of algorithms and computation.

2.**"Computer Organization and Design: The Hardware/Software Interface"**

- **Author**: David A. Patterson, John L. Hennessy
- **Overview**: This reference book is ideal for understanding the design and function of computer systems at the hardware and software interface.

Online Resources:

https://www.coursera.org/
www.udemy.com
www.edx.org

Semester - II

Course Code: APS24BSL107
Course Name: Engineering Mathematics
Course Category: BSC
Credits: 3
Teaching Scheme: L-3
Evaluation Scheme: CA:60, ESE:40
Duration of ESE: 2 Hours
Pre-requisites: Knowledge of matriculation mathematics.
Course Objectives: 1. To Determine range, domain and limit of function. 2. To Apply the fundamental rules of differentiation to various mathematical functions. 3. To Analyze real-world problems and utilize derivative for effective solutions. 4. To Apply the fundamental concepts and techniques of indefinite integration. 5. To Analyze real-world problems and implement integration techniques for effective solutions.
Course Outcomes: Students will be able to achieve & demonstrate the following Cos on completion of course based learning – CO1 – Determine range, domain and limit of function. CO2 – Apply the fundamental rules of differentiation to various mathematical functions. CO3 – Analyze real-world problems and utilize derivative for effective solutions. CO4 – Apply the fundamental concepts and techniques of indefinite integration. CO5 – Analyze real-world problems and implement integration techniques for effective solutions.

Contents:

Unit No.	Content	Teaching Hours
1	**Function and Limit:** 1.1 Definition and representation of function with example. 1.2 Domain, range of function with examples. 1.3 Concept of limit by tangent and velocity problem, and algebraic approach. 1.4 Existence of limit. 1.5 Limit of a function. 1.6 Veridical and horizontal asymptotes. 1.7 Concept of continuity with examples.	10
2	**Differentiation:** 2.1 Concept of derivative with and tangent and rate of change problems. 2.2 Rules of derivative. 2.3 Derivative of functions. 2.4 Chain rule. 2.5 Implicit differentiation. 2.6 Logarithmic differentiation. 2.7 Derivative of inverse trigonometric function. 2.7 Derivative of parametric function.	10
3	**Application of Derivative:** 3.1 Tangent and normal.	10

	3.2 Maxima and minima. 3.3 Monotonicity. 3.4 Roll's theorem, Mean value theorems. 3.5 Linear approximation.	
4	**Indefinite Integration:** 4.1 Concept of integration by area and distance problem. 4.2 Simple Integration: Rules of integration and integration of standard functions. 4.3 Integration by substitution. 4.4 Integration by parts. 4.5 Integration by partial fractions. 4.6 Strategy for integration.	10
5	**Application of Integration:** 5.1 Definite integration. 5.2 Properties of definite integration. 5.3 Area between curves. 5.4 Length of curve. 5.5 Volumes. 5.6 Area of surface of revolution.	10

Text Books:

1 Calculus Early Transcendental, 7e by James Stewart, Cengage Publication.
2 Higher Engineering Mathematics by Grewal B. S.
3 Thomas' Calculus – George B. Thomas, Maurice D. Weir, Joel Hass.
4 Calculus: Early Transcendentals – Howard Anton, IrlBivens, Stephen Davis.

Reference Books:

1. Advance Engineering Mathematics, by Kreysizg, Ervin.
2. Advance Engineering Mathematics, by Das H. K.
3. A Course in Calculus and Real Analysis – Sudhir R. Ghorpade, Balmohan V. Limaye.

ICT / MOOCS: NPTEL / SWAYAM Lectures

Course Code: APS24BSL107
Course Name: Engineering Mathematics Lab
Course Category: BSC
Credits: 1
Teaching Scheme: P-2
Evaluation Scheme: TW:30, PR:20
Pre-requisites: Knowledge of matriculation mathematics and fundamental of computer
Course Objectives: 1. To Determine range, domain and limit of function. 2. To Apply the fundamental rules of differentiation to various mathematical functions. 3. To Analyze real-world problems and utilize derivative for effective solutions. 4. To Apply the fundamental concepts and techniques of indefinite integration. 5. To Analyze real-world problems and implement integration techniques for effective solutions.
Course Outcomes: Students will be able to achieve & demonstrate the following Cos on completion of course based learning – CO1: Determine range, domain and limit of function. CO2: Apply the fundamental rules of differentiation to various mathematical functions. CO3: Analyze real-world problems and utilize derivative for effective solutions. CO4: Apply the fundamental concepts and techniques of indefinite integration. CO5: Analyze real-world problems and implement integration techniques for effective solutions.

Contents –

Sr. No.	List of Experiments	Lab Hours
1.	Introduction to technical tools in mathematics.	2
2.	Visualizing different types of functions and identify their domain and range.	2
3.	Graphical understanding of left and right hand limit.	2
4.	Computation of derivative of different functions.	2
5.	Graphical representation of function and its derivative.	2
6.	Compute tangent and normal to the curve and compute critical points and classify them using the second derivative test.	2
7.	Compute the Integration of basic functions.	2
8.	Graphical representation of Integration.	2
9.	Compute the definite integration of function.	2
10.	Compute the area between two curves and the volume of solids of revolution.	2

Text Books:

1. Calculus Early Transcendental, 7e by James Stewart, Cengage Publication.
2. Thomas' Calculus – George B. Thomas, Maurice D. Weir, Joel Hass.
3. Calculus: Early Transcendentals – Howard Anton, IrlBivens, Stephen Davis.
4. A Course in Calculus and Real Analysis – Sudhir R. Ghorpade, Balmohan V. Limaye.

Reference Books:

1. Python for Data Analysis – Wes McKinney.
2. An Introduction to SageMath – Gregory V. Bard.
3. Exploring Mathematics with GeoGebra –PrzemysławKajetanowicz.

ICT / MOOCS: NPTEL / SWAYAM Lectures

1. GeoGebraResources:https://www.geogebra.org/m
2. Python for Calculus (SymPy):https://docs.sympy.org
3. SageMathDocumentation:https://doc.sagemath.org

Course Code:APS24BSL108
Course Name: Physics - II
Course Category: BSC
Credits: 3
Teaching Scheme: L-3
Evaluation Scheme: CA:60, ESE:40
Duration of ESE: 2 Hours
Pre-requisites: Basic knowledge of mathematics classical mechanics, properties of matter, wave mechanics, and an introduction to nanoscience.
Course Objectives: 1. To Understand and analyse the properties of matter. 2. To Understand the types of motion with proper examples. 3. To Understand concept of wave and it's appropriate examples. 4. To Understand recent application of nanotechnology.
Course Outcomes: At the end of the course, the students will be able to – CO1: Select relevant material in industries by analyzing its physical properties . CO2: Apply laws of motion in various applications. CO3: Apply the concept of SHM and ultrasonic sound for various engineering applications. CO4: Apply the concept of modern Physics for various engineering applications

Contents:

Unit No.	Content	Teaching Hours
1.	**Properties of Matter** 1.1 Deforming Force and Restoring Force, Elasticity, Plasticity, Rigidity. 1.2 Stress and Strain and their types, elastic limit and Hooke's law, types of moduli of elasticity. 1.3 Stress -Strain diagram, Poisson's ratio, factors affecting elasticity 1.4 Defination of pressure, Pressure depth relation, Pascal law, Archimedes' principle 1.5 Viscosity, Viscous Force, Newton's law of viscosity	8
2.	**Kinematics** 2.1 Newton's laws of motion, and their applications. 2.2 Angular displacement, angular velocity, angular acceleration, three equations of angular motion, 2.3 Projectile motion, Trajectory, range of projectile angle of projection ,time of flight 2.4 Work, power and energy: potential energy, kinetic energy, work –energy principle.	7
3.	**Waves and Oscillations** 3.1 Sound waves, amplitude, frequency, time - period, wave-length and velocity of wave, relation between velocity, frequency and time - period of wave. 3.2 Simple Harmonic Motion , Uniform Circular Motion as Simple Harmonic Motion, Equation of simple harmonic motion, Phase of Simple Harmonic Motion. 3.3 Resonance, Application of resonance. 3.4 Ultrasonic waves, properties of ultrasonic waves. 3.5 Piezoelectric and Magnetostriction method to produce ultrasonic waves. 3.6 Applications of ultrasonic waves	8

4.	**LASER and Nanotechnology** 4.1 Planck's hypothesis, properties of photons. 4.2 Photo electric effect: threshold frequency, threshold wavelength, stopping potential, Work function, characteristics of photoelectric effect, Einstein's photoelectric equation 4.3 Photoelectric cell and LDR : principle ,Working and applications 4.5 Laser: properties, absorption, spontaneous and stimulated emission, 4.6 Population inversion, active medium, optical pumping, three energy level system, He-Ne Laser. Engineering applications of Laser. 4.7 Nanotechnology : Properties of nanomaterials (Optical, Magnetic and Dielectric Properties) , Applications of Nanomaterials,	7

Text Books:

1. Physics Textbook Part I - Class XI Narlikar J. V. Joshi, A. W. Mathur, Anuradha
2. Physics Textbook Part II - Class XI Narlikar, J.V. Joshi, A. W. Mathur, Anuradha National Council of Education Research and Training, New Delhi, 2015, ISBN : 8174505660
3. Physics Textbook Part I - Class XII Narlikar J.V. Joshi , A. W. Ghatak A.K. National Council of Education Research and Training, New Delhi, 2013, ISBN : 8174506314
4. Physics Textbook Part II - Class XII Narlikar J.V. Joshi , A. W. Ghatak A.K.

Reference Books:

1. Engineering Physics R.K.Gaur& S.K. Gupta DhanpatRai Publication- New Delhi
2. Applied Physics - II Dr. HussainJeevakhan
3. Engineering PhysicsM N Avadhanulu& P G Kshirsagar S Chand Company Ltd.-New Delhi
4. Fundamental of Physics Jearl Walker "Wiley India Edition

ICT / MOOCS: NPTEL / SWAYAM Lectures

1. https://nptel.ac.in/courses/115106090
2. https://onlinecourses.swayam2.ac.in/nce19_sc05/preview
3. https://iitpal.iitd.ac.in/topic_list/2

Course Code: APS24BSP108
Course Name: Physics - II Lab
Course Category: BSC
Credits: 1
Teaching Scheme: P-2
Evaluation Scheme: TW: 30, PR: 20
Pre-requisites: Basic knowledge of mathematics classical mechanics, properties of matter, wave mechanics, and an introduction to nanoscience.
Course Objectives: 1. To Understand and analyse the properties of matter. 2. To Understand the types of motion with proper examples. 3. To Understand concept of wave and it's appropriate examples. 4. To Understand recent application of nanotechnology.
Course Outcomes: At the end of the course, the students will be able to – CO1: Select relevant material in industries by analyzing its physical properties . CO2: Apply laws of motion in various applications. CO3: Apply the concept of SHM and ultrasonic sound for various engineering applications. CO4: Apply the concept of modern Physics for various engineering applications

Contents –

Sr. No.	List of Experiments	Lab Hours
1.	Determination of Young's modulus of given wire.	2
2.	Comparison of Young's modulii of given materials of wires	2
3.	Determination of relationship between angle of inclination and downward force using inclined plane.	2
4.	Determination of force constant using helical spring.	2
5.	Determination of velocity of sound using resonance tube method.	2
6.	Determination of acceleration due to gravity by using simple pendulum.	2
7.	Determination of I-V characteristics of photoelectric cell.	2
8.	Determination of the divergence of laser beam.	2
9.	Determination of refractive index of glass plate using laser beam.	2
10.	Determination of wavelength of helium neon laser.	2

Text Books:

1. Physics Textbook Part I - Class XI Narlikar J. V. Joshi, A. W. Mathur, Anuradha
2. Physics Textbook Part II - Class XI Narlikar, J.V. Joshi, A. W. Mathur, Anuradha National Council of Education Research and Training, New Delhi, 2015, ISBN : 8174505660
3. Physics Textbook Part I - Class XII Narlikar J.V. Joshi , A. W. Ghatak A.K. National Council of Education Research and Training, New Delhi, 2013, ISBN : 8174506314

4. Physics Textbook Part II - Class XII Narlikar J.V. Joshi , A. W. Ghatak A.K.

Reference Books:

1. Engineering Physics R.K.Gaur& S.K. Gupta DhanpatRai Publication- New Delhi
2. Applied Physics - II Dr. HussainJeevakhan
3. Engineering PhysicsM N Avadhanulu& P G Kshirsagar S Chand Company Ltd.-New Delhi
4. Fundamental of Physics Jearl Walker "Wiley India Edition

ICT / MOOCS: NPTEL / SWAYAM Lectures

1. https://nptel.ac.in/courses/115106090
2. https://onlinecourses.swayam2.ac.in/nce19_sc05/preview
3.https://iitpal.iitd.ac.in/topic_list/2

Course Code:APS24BSL109
Course Name: Chemistry - II
Course Category: BSC
Credits: 2
Teaching Scheme: L-2
Evaluation Scheme: CA:60, ESE:40
Duration of ESE: 2 Hours
Pre-requisites: Basic Engineering Chemistry.
Course Objectives: 1. Describe different processes for manufacturing metals and control of their corrosion 2. Use relevant water treatment process to solve industrial problems. 3. Describe different types of fuels and their properties and uses. 4. Used Relevant engineering materials in Industry
Course Outcomes: At the end of the course, the students will be able to – **CO1:** Understand the process for manufacturing metals and control of their corrosion. **CO2:** Apply water treatment process to solve domestic and industrial problem. **CO3:** Apply the different types of fuel and their uses **CO4**: Understand the engineering material in industry..

Contents:

Unit No.	Content	Teaching Hours
1.	**Metallurgy and Corrosion** **Metallurgy:** 1.1 Introduction 1.2 Definition of metallurgy Properties and application of commercial forms of iron- pig iron ,cast iron and wrought iron **1.3 Steel:** Definition of steel, Preparation of steel from pig iron using open hearth process, basic oxygen process. 1.4 classification of plain carbon steel, low carbon, medium carbon ,high carbon steel with their properties and **Corrosion :** **1.5** Definition, causes of Corrosion Different types of Corrosion and factors affecting corrosion rate. 1.6 Mechanism of corrosion. **1.7** Protection of metal by: Galvanising ,Tinning ,Electroplating 1.8 Paint**:** Definition characteristics, constituent of paint and their function.	08
2.	**Water Treatment** 2.1 Hardness : Classification, Estimation of Hardness by EDTA Method 2.2 Hard water in boilers and prevention: Boiler corrosion, Caustic embrittlement, priming and foaming, scales and sludge's 2.3 Water softening : Zeolite process, Ion exchange process (Cation exchange process and anion exchange) 2.4 Potable water treatment: Characteristics of potable water. 2.5 Potable water treatment: sedimentation, coagulation, filtration, and sterilization. 2.6 Waste water treatment: BOD and COD of sewage water	07

3.	**Fuels and Lubricants** **Fuels:** 3.1 Introduction 3.2 Definition of fuel, calorific and ignition temperature. Characteristic of good fuel 3.3 classification of fuels 3.4 Solid fuels: Coal classification and composition. Analysis of Coal: proximate analysis, Ultimate analysis. 3.6 Liquid fuel: Fractional distillation of crude petroleum, boiling range, composition, properties, application of Bio-diesel 3.7 Gaseous fuel: Composition, properties ,application of Biogas, LPG, CNG **Lubricants:** 3.8 Definition of lubricants .Function of lubricants, 3.9 Classification of lubricants. 3.10 Characteristic : Physical characteristic of lubricants : Viscosity, Viscosity index, oiliness, flash point, fire point, 3.11 chemical characteristic of lubricants- acid value, emulsification, saponification value.(only definition) 3.12 Selection of Lubricants for road rollers, steam engine, sewing machine, concrete mixture, I.C. engine, cutting tools, gears	10
4.	**Non metallic Engineering Material: Polymers** 4.1 Introduction 4.2 Classification of polymer depending upon source ,preparation ,structure, intermolecular force, Polymerisation and its types, Free radical mechancism of addition polymerization 4.3 Plastics and its types thermo softening plastics and Thermosetting plastics 4.4 Preparation , properties and applications of PVC, Bakelite	05

Text Books:

1. Jain P.C & Jain Monica, Engineering Chemistry, DhanpatRai& Sons, Delhi
2. Bhal&Tuli, Text book of Physical Chemistry, S. Chand & Company, New Delhi.
3. O. G. Palanna, Engineering Chemistry, Tata McGraw-Hill Publication, New Delhi.
4. S. S. Dara, A textbook of Engineering Chemistry, McGraw-Hill Publication, New Delhi.
5. ShashiChawla, Engineering Chemistry, DhanpatRai and Co Education and Technical Publishers
6. V. R Gowarikar, Polymer Science, New Age International Ltd.
7. B. K. Sharma, Instrumental Methods of Chemical Analysis, Krishna Prakashan Media

Reference Books:

1. Barrow G.M., Physical Chemistry, McGraw-Hill Publication, New Delhi.
2. ShikhaAgrawal, Engineering Chemistry- Fundamentals and Applications, Cambridge Publishers.
3. Atkins, Physical Chemistry, OUP Oxford.
4. Engineering Chemistry, Wiley India Pvt, Ltd, First editioz

E-Resources:

Name of the website/ E-Journals/ Online Videos

1. NPTEL Basic Courses Engineering Chemistry (https://nptel.ac.in/courses/122/101/122101001/)
2. https://www.ncertbooks.guru/engineering-chemistry/
3. Coursera Chemistry Courses (https://www.coursera.org/browse/physical-science-and-engineering/chemistry?languages=en&page=2)

4. "Introduction to Corrosion" NPTEL Course (https://www.youtube.com/playlist?list=PL8lylDWRkaW8BXestE4baRnN0699S11Lq)
5. "Tribology & Lubrication" NPTEL Course (https://www.youtube.com/playlist?list=PLLwnvFq-JAltJvWAfEVU4gcUK27hEA7FD)
6. "Electrochemistry" NPTEL Course (https://www.youtube.com/playlist?list=PLVFqK_9GOGXnnriQpsn0z1Rss96Rh0vsm)
7. "Modern Instrumental Methods of Analysis" NPTEL Course (https://www.youtube.com/playlist?list=PL400CAFBA72E94CF8)

Course Code: APS24AEP109
Course Name: Chemistry - II Lab
Course Category: BSC
Credits: 1
Teaching Scheme: P-2
Evaluation Scheme: TW: 30, PR: 20
Pre-requisites: Basic Engineering Chemistry.
Course Objectives:
1. Describe different processes for manufacturing metals and control of their corrosion.
2. Use relevant water treatment process to solve industrial problems.
3. Describe different types of fuels and their properties and uses.
4. Used Relevant engineering materials in Industry.
Course Outcomes: At the end of the course, the students will be able to –
LO1:.Design and carry out experiments as well as accurately record and analyse the result of experiments.
LO2: Estimate the impurities present in water using titrimetric and instrumental methods.
LO3: Employ thebasic techniques used in chemistry laboratory for analysis such as volumetric titrations, Complexometrictitration, Viscometer.

Contents –

Sr. No.	List of Experiments	Lab Hours
1.	Prepare the corrosive acidic medium for Aluminium.	2
2.	Determine the rate of corrosion on different temperatures for Aluminium.	2
3.	Determine the percentage of iron present in given Hematite ore by KMnO4 solution.	2
4.	Determine the percentage of copper in given copper ore.	2
5.	Determine Chloride content in given water sample by Mohr's method.	2
6.	Determine total hardness, temporary hardness and permanent hardness of water sample by EDTA method.	2
7.	Determine the alkalinity of given water sample.	2
8.	Determination of acid value of an oil sample.	2
9.	Determine the effect of temperature on viscosity for given lubricating oil using Redwood Viscometer.	2
10.	Determine the moisture and ash content in given coal sample by using proximate analysis.	2

11	Preparation of phenol formaldehyde resin used in manufacturing of Bakelite.	2
12	Prepare Urea formaldehyde resin and understand the structure and properties for its applications in engineering.	2

Reference Books:

1. A Text book on Experiments and Calculations in Engineering Chemistry Author Dr. S. S. Dara, S Chand Publication.
2. Experiments in Engineering Chemistry by Payal B. Joshi, I K International Publishing House Ltd.
3. Systematic Experimental Physical Chemistry by S. W. Rajbhoj& T. K. Chondekar, Anjali Publication.
4. Practical Chemistry with Experiments by Newell, Lymam C.
5. Text Book of Macro and Semi micro inorganic analysis by Vogel. Longman group Ltd London.
6. Engineering Chemistry with Laboratory Experiments by M. S. Kaurav, Asia-Pacific the holdings Private Ltd.

Course Code:APS24VEL110
Course Name: Universal Human Values
Course Category: VEC
Credits: 2
Teaching Scheme: L-2
Evaluation Scheme: CA:30, ESE:20
Duration of ESE: 2 Hours
Pre-requisites: To study the subject of Universal Human Values, there are several key areas of knowledge and understanding that can help build a foundation for the course like Ethics and Moral Philosophy, Human Rights, Philosophy of Life, Cultural and Religious Perspectives, Psychology and Human Behavior, Social Justice and Responsibility ect.
Course Objectives: 1. To create awareness on Engineering Ethics and Human Values. 2. Understand the significance of value inputs in a classroom and start applying them in their life and profession 3. Distinguish between values and skills, happiness and accumulation of physical facilities, the Self and the Body. 4. Understand the role of a human being in ensuring harmony in society and nature. 5. Distinguish between ethical and unethical practices, and start working out the strategy to actualize a harmonious environment wherever they work.
Course Outcomes: At the end of the course, the students will be able to – CO1: Understanding of Fundamental Human Values. CO2: Application of Values in Personal and Professional Life. CO3: Critical Thinking on Ethical Dilemmas. CO4: Promotion of Social Responsibility. CO5: Enhanced Interpersonal Communication. CO6: Cultural Sensitivity and Global Awareness.

Contents:

Unit No.	Content	Teaching Hours
1.	**UHV Introduction** 1.1 Need, Basic Guidelines Content and Process for Value Education, 1.2 Continuous Happiness and Prosperity 1.3 Self-Exploration, Continuous Happiness and Prosperity, 1.4 Right Understanding, Relationship and Physical Facilities.	8
2.	**Harmony of Human Being** 2.1 Harmony in Myself 2.2 Understanding Happiness and Prosperity Correctly. 2.3 Understanding Human Being as a co-existence of the Sentient 'I' and the material 'Body' 2.4 Programs to ensure Sanyam and Swasthya.	7
3.	**Harmony in Human – Human Relationship** 3.1Understanding Harmony in the Family 3.2 Understanding the meaning of Vishwas 3.3 Understanding the meaning of Samman, 3.4 Understanding the Harmony in the Society	8
4.	**Harmony in the Nature and Existence** 4.1 Understanding the Harmony in the Nature 4.2 Natural Acceptance of Human Values	7

	4.3 Visualizing a Universal Harmonious order in Society 4.4 Interconnected ness and Mutual Fulfilment among the Four Orders of Nature. 4.5 Understanding Existence as Co-existence.	

Text Books:

1) CorlissLamont, PhilosophyofHumanism.
2) Gaur.R.R., Sangal.R,Bagaria.
3) G.P,AFoundationCourseinValueEducation,ExcelBooks,2009.Gaur.R.R. ,Sangal. R, Bagaria.G.P,TeachersManualExcelBooks,2009.
4) I.C. Sharma . Ethical Philosophy of India Nagin& co Julundhar.

Reference Books:

1. Mortimer.J.Adler,–What man has made of man.
2. William Lilly Introduction to Ethic Allied Publisher.

ICT / MOOCS: NPTEL / SWAYAM Lectures

E-Resources:-

1. https://www.britishcouncil.in/english/learn-online
2. http://learnenglish.britishcouncil.org/en/content
3. http://www.talkenglish.com
4. Languagelabsystem.com
5. www.wordsworthelt.com

Course Code:APS24ESL111
Course Name: Elements of Mechanical Engineering
Course Category: PCC
Credits: 2
Teaching Scheme: L-2
Evaluation Scheme: CA:60, ESE:40
Duration of ESE: 2 Hours
Pre-requisites: Basic of Physics
Course Objectives: 1. To explain the principles and operation of power plant equipment, including boilers, turbines, and condensers. 2. To analyze industrial material handling systems and their components for efficient mechanical operations. 3. To evaluate the performance and applications of hydraulic pumps, turbines, and related equipment. 4. To demonstrate the working principles of compressors, refrigeration, and air conditioning equipment in thermal systems.
Course Outcomes: At the end of the course, the students will be able to – CO1: Find faults in Thermal Power Plant using acquired knowledge and skills in a given situation. CO2: Diagnose faults of Material handling system using acquired knowledge and skills. CO3: Identify faults of Hydraulic turbines and Hydraulic pumps in a given situation. CO4: Diagnose faults of given Air compressor and Refrigeration system using acquired knowledge and skills.

Contents:

Unit No.	Content	Teaching Hours
1.	**Power plants equipment** 1.1 Layout of Thermal Power Plant, Major thermal power plants in India 1.2 Introduction to steam boilers- BabcockWilcox boilers, Lamont and Loeffler boilers 1.3 Introduction to steam Turbines- Impulse and reaction turbine 1.4 Layout of Portable Generator, Manufacturers and specifications of portable generator 1.5 Introduction to portable generators: I.C engine 1.6 Mechanical parameters measurement Introduction to • Pressure measurement: Bourdon tube pressure gauge • Temperature measurement: Optical pyrometer, Thermocouple • Heat measurement: Calorimeter • Speed measurement of rotating elements: Tachometer, Stroboscope 1.7 Preliminary mechanical faults occurred in steam boilers and turbines	8
2.	**Industrial Material handling systems and components** 2.1 Mechanical components for motion and power transmission: Types and uses of Gears, Belt drives, Chain drives, Bearings, Couplings 2.2 Introduction to material handling systems: Manufacturers, specifications, construction and working of Material transfer lifts, Conveyors, Overhead cranes 2.3 Preliminary mechanical faults occurred in Industrial Material handling	7

	systems	
3.	**Hydraulic pumps, turbines, and Equipment** 3.1 Layout of Hydraulic Power Plant, Major hydraulic power plants in India 3.2 Introduction to hydraulic turbines: construction and working of Pelton wheel, Francis turbine, Kaplan turbine 3.3 Introduction to hydraulic pumps: construction and working centrifugal pump, reciprocation pump and submersible pump 3.4 Preliminary mechanical faults occurred in Centrifugal, reciprocating, and submersible pumps	8
4.	**Compressor, Refrigeration and Air conditioning equipment** 4.1 Introduction to Compressor- Manufacturers, Specifications, construction and working of reciprocating compressor, screw compressor 4.2 Introduction to Refrigeration and Air conditioning : Vapour compression cycle, Construction and working of simple domestic refrigerator and window air conditioner, Manufacturers and specification 4.3 Preliminary mechanical faults occurred in reciprocating compressor and Refrigeration and air conditioning equipment	7

Text Books:

1. P.K.Nag Power Plant Engineering McGraw Hill Education ,ISBN: 978- 9339204044
2. R.K. Rajput Power Plant Engineering Tata-McGraw Hill Education. ISBN : 9788131802557
3. K. Subramanya Hydraulic Machines McGraw Hill Education (India) Private, ISBN, 1259006840, 9781259006845

Reference Books:

1. S.S.Rattan Theory of Machines Tata-McGraw Hill Education. ISBN, 1283187124, 9781283187121
2. C. P. Arora Refrigeration and Air conditioning Tata-McGraw Hill Education ISBN-13: 978-0-07-008390-5
3. Siddhartha Ray Introduction to Materials Handling New Age International Private Limited; ISBN-9788122440072

ICT / MOOCS: NPTEL / SWAYAM Lectures

Course Code:APS24ESP111
Course Name: Elements of Mechanical Engineering Lab
Course Category: PCC
Credits: 1
Teaching Scheme: P-2
Evaluation Scheme: TW:30, PR:20
Duration of ESE: --
Pre-requisites: Basic of Physics
Course Objectives: 1. To explain the principles and operation of power plant equipment, including boilers, turbines, and condensers. 2. To analyze industrial material handling systems and their components for efficient mechanical operations. 3. To evaluate the performance and applications of hydraulic pumps, turbines, and related equipment. 4. To demonstrate the working principles of compressors, refrigeration, and air conditioning equipment in thermal systems.
Course Outcomes: At the end of the course, the students will be able to – CO1: Find faults in Thermal Power Plant using acquired knowledge and skills in a given situation. CO2: Diagnose faults of Material handling system using acquired knowledge and skills. CO3: Identify faults of Hydraulic turbines and Hydraulic pumps in a given situation. CO4: Diagnose faults of given Air compressor and Refrigeration system using acquired knowledge and skills.

Contents –

Sr. No.	List of Experiments	Lab Hours
1.	Identify different components of Thermal Power Plants	2
2.	Demonstrate working of steam turbine	2
3.	Measure temperature of different equipment using temperature measuring devices.	2
4.	Measure pressure of different equipment using pressure measuring devices	2
5.	Calculate Velocity Ratio of given gear/belt drive of suitable mechanical system.	2
6.	Demonstrate working of lift / conveyor used in Industry.	2
7.	Demonstrate Working of Hydraulic Power plant using pelton wheel turbine set up or suitable turbine models /set up	2
8.	Measure pressure, Temperature at different points of Air Compressor.	2

9.	Demonstrate working of household refrigerator for identifying different components and type.	2
10.	Collect information of water lifting systems in ancient India relation with Hydraulic pumps (IKS).	2
11.	Demonstrate working of window air conditioner for identifying different components	2

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1. P.K.Nag Power Plant Engineering McGraw Hill Education ,ISBN: 978- 9339204044
2. R.K. Rajput Power Plant Engineering Tata-McGraw Hill Education. ISBN : 9788131802557
3. K. Subramanya Hydraulic Machines McGraw Hill Education (India) Private, ISBN, 1259006840, 9781259006845

Reference Books:

1. S.S.Rattan Theory of Machines Tata-McGraw Hill Education. ISBN, 1283187124, 9781283187121
2. C. P. Arora Refrigeration and Air conditioning Tata-McGraw Hill Education ISBN-13: 978-0-07-008390-5
3. Siddhartha Ray Introduction to Materials Handling New Age International Private Limited; ISBN-9788122440072

ICT / MOOCS: NPTEL / SWAYAM Lectures

Online Resources:

1. **https://www.tutorialspoint.com**
2. **https://javatpoint.com**
3. **https://www.w3schools.com**

Course Code: APS24VSL112

Course Name: Web Development Lab

Course Category: VSEC
Credits: 2
Teaching Scheme: P-2
Evaluation Scheme: TW:60, PR:40
Pre-requisites:
Course Outcomes: At the end of the session, the students will be able to -
CO1: Use of website and content of website designing in web page.
CO2: Use block level and text level formatting tags to present content in web page.
CO3: Arrange list and apply hyperlink on images.
CO4: Organize and arrange content using table and frames.
CO5: Apply presentation schemes on content of web page using CSS.
CO6: Publish website on Internet or Intranet.

List of Experiments–

Sr.No.	List of Experiments	Practical Hours
1	Install any one editor and display hello message on screen.	2
2	Create a web page using Block Level tags and HR tags.	2
3	Create a web page using Text Level tags and display special character in web page.	2
4	Create a web page for implementation of different types of list.	2
5	Create a web page for inserting image in web page and change attribute of image.	2
6	Create a web page to link: a) Different web page of same web site b) Different web page of different web site c) Images	2
7	Create a web page for implementation of table tags	2
8	Create a web page for implementation of frame tags	2
9	Create a web page by implementation of different selector	2
10	Create a web page by using different CSS properties	2
11	Create a web page to create different forms: a) Login form b) Student Entry From c) Forms with linking of different tags	2
12	Create above web page and connect with database	2
13	Create a web page for multimedia a) Audio b) Video c) Source	2
14	Install a web server for publishing website	2
15	Publish a web site on internet by using free hosting site.	2
16	Mini Project	6

Course Code:APS24VSP113
Course Name: Engineering Workshop - I

Credits: 2
Teaching Scheme: P-4
Evaluation Scheme: TW:60, PR:40
Pre-requisites: Basic Knowledge of fundamentals of applied science and material science
Lab Outcomes: At the end of the course, the students will be able to –
LO1: - Use firefighting tools and equipment.
LO2: Prepare job using different tools in fitting shop.
LO3:. Perform various operations using plumbing and carpentry tools.
LO4: Prepare various welding joints.
LO5: Produce simple job using different sheet metal operations.

Contents:

Sr. No.	Name of Experiment	Practical Hours
1.	School within School: Each student will demonstrate and explain at least one tool (to be assigned by teacher) to all batch colleagues. ii: Each student will share his/her student activities outcome. He/she will also share the experience for the student activities he/she has carried out.	2
2.	Identify fire extinguisher according to their specification.	2
3.	Perform mock drill session in group of minimum 8 students for extinguishing fire.	2
4.	Prepare one simple and another male-female type fitting jobs as per given drawings- 2 job.	8
5.	Familiarize students with the interface and basic functions of the welding simulator.	2
6.	Learn welding techniques for various positions (flat, horizontal, vertical, overhead).	6
7.	Prepare utility job (like stool, benches, tables or similar jobs) involving arc welding and artificial wood as per given drawing (in group of 4 to 5 students) Fabrication operation involve measuring, marking, cutting, edge preparation, welding	8
8.	Prepare utility job (like stool, benches, tables or similar jobs) involving wood working as per given drawing (in group of 4 to 5 students) Fabrication operation involve measuring, marking, cutting, edge preparation, welding.	6
9.	Prepare sheet metal utility job using following operations a. Cutting And Bending b. Edging c. End curling d. Lancing e. Soldering f. Riveting.	6
10.	Prepare turning utility job using following operations a. Cutting And turning b. facing c. knurling d. threading e. Boring.	6